

Institute of Actuaries of India

Subject ST1 – Health and Care Insurance Specialist Technical

September 2016 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**(i)**

For all assumptions, a prudent basis with appropriate margins for adverse deviations should be used rather than best estimate

Regulatory requirements should be taken into account

a) Morbidity

- Need to consider both claim incidence and claim recovery rates
- Need to consider if the rates for those who have taken up the CI rider should be higher
- Allow for RPI – a higher rate is prudent
- Pre Claim
 - Company has lot of experience – have regard to company's own past experience
 - Look at by age, gender, occupation class, type of industry and territory
 - Have regard to underwriting policy and changes over time
 - Have regard to changes in economic climate (more claims expected when there is economic downturn)
 - A high incidence rate is prudent
- Post Claim
 - Have regard to definition of disability
 - Have regard to company's own past experience by risk factors as for pre-claim
 - Duration of claim
 - Have regard to replacement ratio implied by the income chosen
 - A low recovery rate is prudent
- CI incidence
 - Need CI incidence assumption for policies that have taken up the rider
 - Have regard to age, gender and size of cover
 - Regard to published tables and industry experience
 - And, allow for the guaranteed issue without fresh underwriting
 - A higher CI incidence rate is prudent

b) Expenses

- Should allow for maintenance as well as claim expenses
- Should be based on analysis of company's recent experience
- Need to allow for future expense inflation
- Have regard to claim management practices

- Have regard to territory as the costs could vary
- Have regard to legal costs and trends
- Allow for renewal commission to intermediaries

c) Valuation rate of interest

- Regard to currency
- Regard to yields on existing assets
- Regard to yield on sums to be invested in the future
- Credit/default risk
- Regard to term of liabilities
- A lower investment return assumption is prudent

Valuation rate of interest, expense inflation and RPI should all be consistent

[12]

(ii) The reasons for higher investment return assumption:

- Excellent investment management team - has been earning higher yields on the assets backing the liabilities consistently
- large asset under management and so able to dynamically move across asset classes to take advantage of investment opportunities
- lot of free assets and so able to execute riskier investment strategy yielding much higher returns
- excellent asset and liability matching – so, lower margin for adverse deviation and reinvestment risk
- Large and established player and so the regulatory investment restrictions are more relaxed compared to other smaller players
- has been able to take advantage from currency exchange movements
- Liability is linked to actual RPI while others may have fixed inflationary increase – so, can be more aggressive with investment approach to generate higher returns

[3]

(iii) Possible reasons for low take up rates of CI rider:

- Lack awareness or forgotten about the CI option available
- CI rider is expensive and so only those whose health has deteriorated and expect to make a claim take-up the rider
- ..and those healthy ones have cheaper alternatives
- No additional commission/incentive paid to the sales force for the CI rider
- No active promotion

- Income protection is perceived to take care of the needs and so CI rider is not seen required
- A competitor has launched an attractively priced CI with more illnesses covered
-and accepts policyholders of other insurers who were underwritten in the last 5 years for life or health products without further underwriting
- CI rate are not guaranteed

The ways by which the Company may increase the take-up rate:

- Incentives to the sales force
- Run a focused campaign
- Offer price reduction to those willing to undergo underwriting
- Remind the policyholders' of the option at the end of 5 years
- add more illnesses to the rider
- ...and/or introduce return of premium on no claim

[6]

[21 Marks]

Solution 2:

(i) Typical features of a PMI product:

- indemnity product that seeks to provide compensation for the cost of private medical treatment
- provides cover for acute rather than chronic conditions
- usually the insurer pays the treatment provider directly
- yearly renewable – premium can vary from year to year
- requires minimum period of hospitalization
- mostly in-patient only but may cover out-patient treatment to some extent
- Some form of co-insurance or sub-limits to prevent overuse
- Excludes pre-existing conditions, GP services (ie visits to your doctor) and cosmetic treatments
- Many policies now cover types of treatment such as homeopathy and acupuncture
- General exclusions: Drug abuse, self-inflicted injuries, HIV/AIDS, hazardous pursuits etc.

[4]

(ii) Pricing assumptions, data and their sources

- claim incidence rate and average claim cost
- ...by age, gender, sum assured and territory
- ...by occupation/industry class, especially for the Group PMI
- ...experience from hospital cash and CI products can be used
- ... allowing for any differences in the target market and underwriting
- ...regard to industry tables but allow for any product and underwriting differences
- ...regard to reinsurance rates
- ...regard to legal trends from outcome of recent court cases
- Medical trend (to allow for trend in utilization and medical cost inflation)
- ...industry experience and advice from consultants/reinsurers
- ... and have regard to any medical infrastructure developments and Govt. policies with respect to health care provisions
- expense and expense inflation
- ...recent experience analysis
- ... cost of TPA could be a good indicator of claim administration cost
- ...benchmark with published studies
- ...regard to expected volume and persistency to spread the cost
- ...based on experience of other providers in the market and the likely impact of the special features
- Interest rate (not a significant assumption though)
- ...yields realized on short-term investments
- ...consistent with expense inflation
- Incidence rate for emergency transportation and cost
- ...incidence rates from hospital cash and CI products
- ...cost of medical transportation from the providers of such services
- ... regard to the fact that the cost may increase if this product increases demand for such services
- ...regard to any published tables on major accidents
- Incidence and cost of second opinion
- ...most major illness will go for second opinion
- ...carve out from the experience of hospital cash and CI products
- ...experience of any overseas products with such a feature
- ...reinsurers might be able to provide data
- ...regard to availability/access – will vary by territory
- ...cost of OP consultation with specialist doctors will be a good indicator of the cost
- Incidence and cost of overseas treatment

-look at incidence of major illnesses for which the local treatments not available from hospital cash and CI products
- ...published studies on people going overseas for treatment and discount those who overseas for perceived high quality
- ...cost overseas based on experience of any international PMI products (reinsurers might have access to this data)
- data from providers of medical transportation services

[11]

(iii)

- New product and additional new features – gives rise to higher degree of mis-pricing risk
- Benefit payments are determined by individuals over whom the company has little or no control (eg. Increase in fee for medical transportation services, cost of treatment etc.)
- Claim frequency is a function of access to care and medical advancements that are difficult to predict
- Greater uncertainty surrounding the incidence and cost of new features with no direct local experience available
- risk of a single large claim (eg. overseas treatment)
- ... or a large number of claims arising from a single incident (especially, the group PMI)
- These risks can be mitigated through appropriate reinsurance protection
- ...and negotiated tariffs for various types of medical treatment with a network of hospitals
- ...and reviewable premium rate structure and so the pricing can be adjusted with experience
- ...and no claim discount to discourage over-claiming
-introduction of co-pay or deductible to discourage small claims
- ...significant co-pay and an appropriate upper limit for overseas treatment to mitigate misuse
- ...underwrite to prevent anti-selection and over-insurance
- Any enhancement to State provision of health care will increase the risk of poor sales leading to inability to cover expense and potentially insolvency
- ...ability to react quickly to such changes by way of changes to the product features will help mitigate the risk to some extent
- Lack of experience with indemnity products increases the reputation risk caused by more disgruntled policyholders – this will affect the sale of other products too

- ...claim administration through a professional TPA and claim pre-authorization requirement will help mitigate the risk to some extent

[7]

[22 Marks]

Solution 3:

(i)

- Type of Products and their features e.g. PMI, Hospital Cash.
- Sales volume projections for each of the 5 years split by product
- Initial Capital to be deployed
- Outsourcing arrangements (eg. Customers help desk, TPA for claims)
- Information on how the existing infrastructure/expertise of the partners will be leveraged
- Investment strategy
- Break-even expectation of the shareholders
- Return on capital expectation of the shareholders
- reinsurance arrangements

[3]

(ii) Products and Business Volume:

- The growth in premium volumes should be consistent with the current prevailing market condition
- Model points that closely reflect the expected business profile
- Fit with overall business plan, company culture and strategy

Expenses

- Cost of setting up the online platform and ongoing maintenance
- Number of operations staff required and their salary– consistent with the volume projections
- Cost of TPA , investment management and other outsourcing costs
- Property and other infrastructure related cost
- ... since the distribution is online a few centralized units would be required as opposed to offices spread across the country
- ... will also depend upon how much of underwriting and claim administration can be online

Investment Return

- Project the expected asset mix (with minimum consideration to regulatory limits)
- What is the expected yield on each asset class?
- Is expected yield flat in all years? This should allow for any expected downturn in the interest rate.
- Calculate the weighted average yield on the portfolio.
- Investment return assumption may vary by product/line of business.
- Investment return assumption should be consistent with expense inflation assumption

Mortality and Morbidity

- Mortality and Morbidity assumptions would be based on the market trend
- ... if market study is not available, then reinsurers can help
- Regard to expected underwriting policy
- ...since it is only online distribution there may limitation on the level of underwriting
- These assumptions may vary by product, age, sex and pre/post claim stage.
- Consider duration of claim, number and amount of claim and claim volatility over the years to arrive at Mortality and Morbidity Assumptions.
- These assumptions may be different in pre claim and post claim period

Expense

- Need assumptions for acquisition (deferrable), acquisition (non-deferrable), maintenance and overhead.
- Need split by fixed and variable – to determine flat, per claim, % of premium, % of sum assured etc.
- ...the structure and the level may vary by product.
- expense inflation – consistent with the interest rate assumption
- benchmark the assumptions against the experience of similar new insurers in the market

Persistency

- By product, by policy size, by duration and by territory
- Benchmark with experience of online distribution channel of similar insurers
- Need to be set consistent with the premium
- ...more competitive than the market would imply higher persistency
- Need to be consistent with the expense assumption
- ...higher persistency would mean more policies to spread the fixed expenses over

Others

- Expected reinsurance terms and their impact on cash-flows
- All pricing assumptions will be best estimate
- ...and the reserving will need to be prudent with margins for adverse deviations
- have regard to regulatory restrictions on investment, premium reviewability, pricing margins
- do reasonableness checks on the projected financial outcomes based on your experience in the market

[11]**(iii)**

- Project available solvency margin (excess of assets over liabilities)
- regard to professional guidance
- exclude inadmissible assets as per regulatory norms
- project required solvency margin as per regulatory requirements
- stress test to ensure the solvency is maintained in a range of possible scenarios
- ..including business volume (to reflect new business strain), claim experience etc.
- solvency Ratio (available solvency margin / required solvency margin) should be at least equal to the minimum prescribed by the regulator
- ..shareholders may like to maintain a ratio more than the minimum required
- a lower than required level of solvency ratio would require additional capital provision

[3]**[17 Marks]****Solution 4:****(i)**

- First analyse the basis of CFO's concern – is that excess of reinsurance premium over reinsurance claim recovery?
- Check that the premiums and claims do correspond – eg. earned premium vs incurred claims
- Analyse the reinsurance outgo vs reinsurance claim recovery – is it across the board?
-Particular product(s)? particular rating factor(s) - gender, occupation class, territory etc.?
- ...it may be that there may be some cross-subsidy – some parts the reinsurer is losing and in others gaining and overall gaining
- Cross-check the reinsurance financials with the reinsurer so that there is no data or reporting issues

- ...need to make sure that there is no overpayment of reinsurance premium and that there will be refund from the reinsurer in the future
- ...similarly, there is no underpayment – that would mean the reinsurer's profit is more than what we are seeing now
- ..check if there is any experience refund arrangement and, if so, that it is taken into account in the analysis
- ..check if the experience is sufficiently credible
- Check if there has there been any reduction in the reinsurance rates and so the reinsurance financials will change from what has been the case
- For each product, need to determine the appropriate structure and retention taking into account
- ...QS is more appropriate when the concern on the claim incidence rates and not so much on the size of claim
- ...and surplus arrangement will be more appropriate if it the concern is the other way round
- ... and a combination of both will need to be considered as well depending upon the risk exposure
- ...concerns of accumulation, catastrophe would require non-proportional arrangements such as XoL
- Vary the retention levels and calculate the probability of insolvency
- ...a stochastic model could be used with the claim rates as stochastic variable
- Project expected claims along with the company's assets and liabilities
- Use simulation techniques to compute the retention level so that the company stays solvent, say 995 out of 1000 runs
- Alternatively, vary the retention level such that the cost of financing a risk experience fluctuation reserve and cost of financing is minimized
- Reinsurance rates tend to vary with retention levels
- ...so, compare reinsurance premiums to claims at each retention level allowing for the different rates available
- ..consider if more than one reinsurer will need to used
- ..depends on the size of business and specialization of the reinsurer

[9]

(ii)

- Relationship with the reinsurer(s) – how easy or difficult to work with?
- Technical assistance with underwriting, claims
- ...product innovation and development
- Provision of IT systems for underwriting, claim management, reinsurance administration

- Financial strength of the reinsurer(s)
- Business growth strategy of the company and which reinsurer(s)/ structure/level of reinsurance would best fit in
- ..for example, expansion into a new territory
- Diversification of counter party exposure (do we need to involve more than one reinsurer?)
- Does the reinsurer offer any 'financing' in some form that will help manage new business strain?
- Does the reinsurer help with other aspects of business...
- ..evaluating a TPA, tie-up with any specialist providers or distributors
- ..quality control audits, regulatory filing etc.

[4]

[13 Marks]

Solution 5:

(i)

- The main objective is that the design should meet that needs of the customer
- Should reasonable cover care fees including any assistive devices in an institutional care facility until death
- ...when the insured is totally and permanently disabled and
- ...unable to perform activities of daily living even with assistive devices
- includes an inflation-proofing at a flat rate
- ...full inflation-proofing would be prohibitively expense
- Individual unit-linked product platform used...
- and so there is flexibility with contributions (premium), and
- ..easier to review rates as experience unfolds
- Policyholders will be advised of the level of contribution required from time to time to earn adequate points to meet the cost of care
- Policyholders earn benefit points in proportion to the contribution that they make
- ... and then applicable charge for the cost of care and their underwriting status
- ... the charge will inflate at the same rate as the flat benefit inflation
- ...underwriting will group risks broadly based on their current level of disability and its likely impact on the need for long term care
- Investment returns are also converted into benefit points in proportion to the contributions
- A small mortality benefit to cover the funeral cost is included and indexed at the same rate
- ...mortality charge will be deducted from the contributions/unit-linked fund
- Govt. contributes an amount equal to the individual's contribution

- The benefit points accumulated will be used to purchase an immediate needs indexed annuity at the time of needing care (TPD on ADL basis) at the then rates

[8]

(ii)

- Investment strategy should have regard to the nature, term and currency of the liabilities
- ...and regulatory restrictions
- ...and that there is enough diversification
- Govt. securities is a very safe and very liquid but tend to yield low returns
- ..however, they do not provide good match to the nature of liabilities unless the securities are inflation-linked
- Equities, on the other hand, is risky but provides good inflation-proofing
- ... some sort of life-staged investment strategy will be more effective
-start off with greater exposure to equities, index-linked bonds, and to some extent property and
- ...as the benefit payout stage approaches move them toward secure and liquid investments
- to avoid any shock from sharp fall in the equity market when the investments have to be realised
- While the product is unit-linked with the investment risk being passed on to the policyholder...
- ...the investment strategy should aim to maximize returns with an acceptable level of risk
- Govt. may stipulate certain investment approach on the subsidy that it provides

[4]

[12 Marks]

Solution 6:**(i)**

- Individual PMI – each individual is underwritten.
- ...proposal form to be filled in and medical test depending upon the age and the amount of cover
- ...each individual will therefore be either accepted at standard terms, charged extra or declined cover
- Group PMI – typically no individual underwriting
- ... there is underwriting of the group based on its past experience
- ...and other risk factors such as territory, level of exposure to occupational hazards,

- ...level of participation, distribution of cover (are there some lives with very high cover?) etc.
- Once a group is acceptable from an underwriting perspective a free (of underwriting) cover limit (FCL) is offered to the group
- .. all members of the group is covered up the FCL
- Individuals with cover beyond the FCL are individually underwritten
- ...however, all of those lives will also get cover up to the FCL regardless of the underwriting outcome
- Groups with voluntary participation and low expected take-up will tend to be treated more like individual PMI from an underwriting perspective

[4]

(ii)

- Brokers are independent of a particular insurance company
- Usually remunerated through commission from the company or a fee from the client
- Usually client initiates sales although the brokers may market to potential clients
- Able to sell complicated products and offer market comparisons
- Typically deal with high net worth individuals or corporates
- Brokers can leverage their existing relationship with corporates (eg. already offering general insurance solutions to the corporate)
- Brokers do provide insurance administration support to the clients
- Corporates do want an independent advice to the insurance purchase process so that they get value for money
- The current sales force, tied agents, may not be seen independent and so may not be preferable by the corporates
- For these reasons, brokers could be seen an appropriate channel to distribute the Group PMI product
- Brokers, on the other hand, tend to churn the business (move the business from one insurer to another) in order to maximize their remuneration
- ...this depends upon the market practice on broker remuneration (this may be dictated by regulation)
- Some large corporates may like to deal with insurance companies directly
- ...especially, they have an insurance department and has lot of experience dealing with insurance companies and like to avoid fee to the broker
- ..so, it worth considering a dedicated direct sales force to distribute group PMI
- To the extent the Group PMI is a market standard product, it may be possible to offer the product online directly to the potential clients

[6]

(iii)

- There is deliberate cross-subsidy between individual and group products by the insurers
- ...premiums are set such a way that individual policies subsidize group
- ...the insurers focus on profitability of the portfolio (individual + group) overall
- ...greater negotiation power by group clients due to large amount of premium involved
- Group is heavily broker driven and so very competitive
- ...reinsurers do not support cross-subsidy and so bulk of the group risk is retained by the insurers
- ...more reinsurance for individual – and so greater pricing discipline than group
- Remuneration structure for brokers encourages churning of business
- ...so, insurers undercut price on renewals to retain business in order to cover upfront costs incurred
- The market has recently opened up for private participation from Govt. owned monopoly insurer – so all players are looking at building scale – top line focus
- Market leadership is measured by the topline and group is quick to build up top-line given the size of premium involved
- Insurers' want to acquire customers though Group PMI products...
- ...and use the data base to upsell/cross-sell other products
- ...and use claim information to underwrite the group members to offer other more profitable products on a pre-underwritten basis
- Increase in the incidence of catastrophic events (eg. Pandemics, workplace accidents etc.) leading to large losses for group..
- ... the impact less pronounced for individual as they are quite well spread..

[5]

[15 Marks]
